# Post-Quantum Cryptography Migration

## Part 15 - Where It Runs: Email & Messaging

**Compiled by the Spaced Research Ledger**

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Messages are the purest harvest-now-decrypt-later target: personal, durable, and often archived by the recipient for years. That has made the secure messengers the most advanced post-quantum deployments anywhere, while email, the older medium, mostly waits on the layers beneath it.

This report covers both worlds. Section 1 covers email security, from transport policy to encrypted mail standards. Section 2 covers the secure messaging protocols, where the state of the art now lives.

Email's milestone is a standard, not a deployment. RFC 9980 brought post-quantum algorithms into OpenPGP in June 2026, defining composite encryption and signatures. The immediate complication is that GnuPG, the most widely deployed implementation, follows an incompatible specification instead.

Messaging's milestone is engineering. Signal extended post-quantum protection from session setup into the ongoing message ratchet itself, Apple's iMessage has run continuous post-quantum rekeying since 2024, and both protocols carry machine-checked security proofs. Nowhere else in this collection is the migration this far along.

## 1. Email Security

*Email security is layered: transport policies decide whether mail travels over TLS at all, and end-to-end standards like OpenPGP and S/MIME encrypt the message itself. The transport layers inherit their quantum future from TLS and DNSSEC. The message formats needed new standards, and 2026 delivered them.*

### Recent Developments

OpenPGP crossed the standards line. RFC 9980 published in June as a Standards Track extension defining post-quantum public-key algorithms for the protocol [1]. It specifies composite encryption pairing ML-KEM with elliptic curves, composite signatures pairing ML-DSA with EdDSA, and standalone SLH-DSA [1]. The immediate fracture: GnuPG does not implement it, following the incompatible LibrePGP specification instead [2].

S/MIME's machinery is close behind. The composite-KEM conventions for CMS entered Last Call [3], and the underlying composite draft keeps revising [4]. A formal analysis showed the sharp edge: a message encrypted to many recipients is only post-quantum confidential if every path to the content key is post-quantum [5]. A validation framework is proposed to check exactly that [5].

The transport policies, by design, have no post-quantum extensions of their own. MTA-STS inherits from the TLS stack [6], and DANE inherits from DNSSEC signing plus TLS [7].

### Current State

The transport layer is a deployment story, not a cryptography one. DANE authenticates mail servers through DNS records instead of certificate authorities [7], and depends entirely on DNSSEC integrity [8]. Gmail validates it and Postfix has long supported it [9], with Microsoft's inbound support arriving in 2024 [10]. Adoption concentrates in Europe [8], monitored through the standard TLS reporting mechanism [9].

MTA-STS, standardized in 2018 [6], remains rare: under 2 percent of analyzed U.S. domains implement it against near-universal adoption of the sender-authentication standards [11], and under 1 percent of the top million domains, though growth has doubled in two years [12]. Deployment practice runs testing mode before enforcement [13].

OpenPGP's post-quantum work is reaching real software. The specification makes X25519 mandatory for the composite combinations [14]. The project began at the IETF in 2022 [15], and implementations are underway in Thunderbird under German government funding and in Proton's open-source libraries [15]. The multi-recipient caveat applies here too: confidentiality is only post-quantum when every recipient key supports it [1].

S/MIME's composite path is built for insurance. Composite ML-KEM targets exactly the applications using X.509 and CMS that want protection against a catastrophic ML-KEM break [16], with strict key-usage separation [17]. The plain path is already ratified: RFC 9936 specifies ML-KEM in CMS through the standard recipient mechanism [18].

One healthcare-specific ecosystem now sits on this map with nothing to show. Direct Secure Messaging, the certificate-anchored network US healthcare uses to move sensitive records, encrypts with digital certificates and a public key infrastructure [40]. That is exactly the kind of PKI a post-quantum migration eventually reaches. Its steward, DirectTrust, has published no post-quantum migration artifact at all: no certificate-policy update, no standards work item, a gap confirmed by repeated targeted searching. The artifact to watch for is a certificate policy or trust-anchor update naming the NIST algorithms.

## 2. Secure Messaging Protocols

*The secure messengers invented modern end-to-end encryption, and they are inventing its post-quantum form too. The problem is harder than TLS: sessions last years, devices go offline, and every key update must survive lost messages. This section covers Signal's new ratchet, Apple's PQ3, and the group-messaging standard behind enterprise chat.*

### Recent Developments

The group-messaging standard tightened its options. The MLS post-quantum cipher-suite draft dropped its X-Wing hybrid options, focusing on ML-KEM hybrids and pure ML-KEM suites [19].

### Current State

Signal moved the frontier twice. Its PQXDH key agreement, deployed since 2023, computes session secrets from both X25519 and Kyber so an attacker must break both [20]. It is formally analyzed with tools that refined the specification itself [21], stands at its third revision [22], and runs in every current client [20]. Its authentication remains classical for now [23]. The 2025 step is the Sparse Post-Quantum Ratchet, mixed with the existing Double Ratchet into what Signal calls the Triple Ratchet [24].

It extends post-quantum protection to the ongoing message lifecycle using ML-KEM-768 [25] and rolls out transparently with backward compatibility [26]. The kilobyte-plus key material travels through chunking and erasure coding [25]. The design was built and verified with outside researchers [25], the specification now documents the combined protocol [27], and Signal plans eventually to force all conversations onto it [28].

Apple’s PQ3 got there earlier on one axis. It combines Kyber-derived keys with elliptic curves and extends a ratchet for post-quantum post-compromise security [29], rolling out with the early-2024 OS releases [29]. Apple classes it at its highest security level, covering initial establishment and ongoing exchange [29]. It mixed fresh post-quantum keys into derivation every fifty epochs, ongoing rekeying that predated Signal’s equivalent [30], and carries machine-checked proofs over both ratchets [31].

The tradeoffs are documented. There is no cryptographic deniability, since every message is signed [32]. Most exchanges still use classical key agreement by default, with full post-quantum exchanges triggered heuristically to save bandwidth [32].

The rest of the field inherits or waits. WhatsApp builds on the Signal Protocol [33] and inherits its post-quantum upgrades rather than engineering its own [34], inside Meta’s company-wide migration framework [35]. Matrix plans to resume its post-quantum work in November [36]. MLS, the IETF’s group-messaging standard, has registered post-quantum cipher suites in draft [37].

They provide post-quantum confidentiality while most suites keep classical signatures [37], and remain an active draft rather than an RFC [37]. A companion draft amortizes the cost by combining a classical session with a post-quantum one [38], superseding the expired pre-standard hybrid [39].

## About This Report

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## Sources


1. RFC 9980 - Post-Quantum Cryptography in OpenPGP - https://datatracker.ietf.org/doc/rfc9980/ - as of 2026-06-01
2. Post-Quantum OpenPGP: RFC 9980 Standard Explained - https://postquantum.com/security-pqc/post-quantum-openpgp-rfc-9980/ - as of 2026-07-05
3. draft-ietf-lamps-cms-composite-kem-01 - https://datatracker.ietf.org/doc/draft-ietf-lamps-cms-composite-kem/ - as of 2026-08-06
4. draft-ietf-lamps-pq-composite-kem-19 - Composite ML-KEM for use in X.509 Public Key Infrastructure - https://datatracker.ietf.org/doc/draft-ietf-lamps-pq-composite-kem/ - as of 2026-08-14
5. Analysing the Post-Quantum Security of S/MIME - https://eprint.iacr.org/2026/1374 - as of 2026-07-04
6. RFC 8461: SMTP MTA Strict Transport Security (MTA-STS) - https://www.rfc-editor.org/rfc/rfc8461.html - as of 2026-08-15
7. How SMTP DNS-based Authentication of Named Entities (DANE) works | Microsoft Learn - https://learn.microsoft.com/en-us/purview/how-smtp-dane-works - as of 2026-08-15
8. What is DANE? DNS-Based Authentication of Named Entities Explained (2026) - https://securityboulevard.com/2026/04/what-is-dane-dns-based-authentication-of-named-entities-explained-2026/
9. DANE/TLSA: the complete guide to DNS-based email certificate authentication - https://www.captaindns.com/en/blog/dane-tlsa-complete-guide
10. DANE for Microsoft 365 and Exchange Online: complete guide - https://www.captaindns.com/en/blog/dane-tlsa-microsoft-365-exchange - as of 2024-03-01
11. PowerDMARC Releases United States DMARC & MTA-STS Adoption Report 2026 - https://www.barchart.com/story/news/37303214/powerdmarc-releases-united-states-dmarc-mta-sts-adoption-report-2026 - as of 2026-01-29
12. MTA-STS adoption: three years of steady growth - https://www.uriports.com/blog/mta-sts-survey-update-2026/ - as of 2026-01-06
13. Learn about MTA-STS and TLS - https://redsift.com/guides/email-protocol-configuration-guide/learn-more-about-mta-sts-and-tls
14. Post-Quantum cryptography in OpenPGP NIST Fifth PQC Standardization Conference - https://www.wussler.it/NIST5thPQC.pdf
15. Post-Quantum Cryptography in OpenPGP - FOSDEM 2025 - https://archive.fosdem.org/2025/events/attachments/fosdem-2025-5992-post-quantum-cryptography-in-openpgp/slides/238257/PQC_in_Op_AcmguXM.pdf - as of 2022-03-01
16. draft-ietf-lamps-pq-composite-kem-06 - https://datatracker.ietf.org/doc/html/draft-ietf-lamps-pq-composite-kem-06
17. draft-ietf-lamps-pq-composite-kem-06 - Composite ML-KEM for use in X.509 Public Key Infrastructure and CMS - https://datatracker.ietf.org/doc/draft-ietf-lamps-pq-composite-kem/06/
18. RFC 9936 - Use of ML-KEM in the Cryptographic Message Syntax (CMS) - https://datatracker.ietf.org/doc/html/rfc9936 - as of 2026-03-01

19. draft-ietf-mls-pq-ciphersuites-06 - ML-KEM and Hybrid Cipher Suites for Messaging Layer Security - https://datatracker.ietf.org/doc/draft-ietf-mls-pq-ciphersuites/ - as of 2026-07-21
20. Signal >> Blog >> Quantum Resistance and the Signal Protocol - https://signal.org/blog/pqxdh/ - as of 2023-09-19
21. Cryspen | An Analysis of Signal's PQXDH - https://cryspen.com/post/pqxdh/ - as of 2023-10-20
22. PQXDH - Signal Wiki - https://signal.miraheze.org/wiki/PQXDH - as of 2024-01-23
23. Signal >> Specifications >> The PQXDH Key Agreement Protocol - https://signal.org/docs/specifications/pqxdh/ - as of 2024-01-23
24. Signal >> Blog >> Signal Protocol and Post-Quantum Ratchets - https://signal.org/blog/spqr/ - as of 2025-10-02
25. Signal Adds Post-Quantum "Triple Ratchet" Protocol for Stronger Security - https://cyberinsider.com/signal-adds-post-quantum-triple-ratchet-protocol-for-stronger-security/
26. Signal Implements Hybrid Post-Quantum Ratchet to Mitigate Quantum Computing Risks - https://cyberpress.org/signal-implements-hybrid/
27. Signal >> Specifications >> The Double Ratchet Algorithm - https://signal.org/docs/specifications/doubleratchet
28. Signal Adds Post-Quantum "Triple Ratchet" Protocol for Stronger Security - https://cyberinsider.com/signal-adds-post-quantum-triple-ratchet-protocol-for-stronger-security/ - as of 2025-10-03
29. iMessage with PQ3: The new state of the art in quantum-secure messaging at scale - Apple Security Research - https://security.apple.com/blog/imessage-pq3/ - as of 2024-02-21
30. Apple's New iMessage, Signal, & Post-Quantum Crypto | CSA - https://cloudsecurityalliance.org/blog/2024/05/17/apple-s-new-imessage-signal-and-post-quantum-cryptography
31. A Formal Analysis of Apple's iMessage PQ3 Protocol - https://eprint.iacr.org/2024/1395
32. Post-quantum messaging: examining Apple's new PQ3 protocol | PQShield - https://pqshield.com/post-quantum-messaging-examining-apples-new-pq3-protocol/
33. A Post-Quantum End-to-End Encryption Protocol | IEEE Conference Publication | IEEE Xplore - https://ieeexplore.ieee.org/document/10469296/
34. Is WhatsApp Prepared for Post-Quantum Security? | Post-Quantum Security Center - https://www.gopher.security/post-quantum/is-whatsapp-prepared-for-post-quantum-security
35. Post-Quantum Cryptography Migration at Meta: Framework, Lessons, and Takeaways - https://engineering.fb.com/2026/04/16/security/post-quantum-cryptography-migration-at-meta-framework-lessons-and-takeaways/ - as of 2026-04-16
36. Matrix slides (2025) - https://2025.matrix.org/slides/slides_8PEMJB.pdf
37. ML-KEM and Hybrid Cipher Suites for Messaging Layer Security - https://messaginglayersecurity.rocks/mls-pq-ciphersuites/draft-ietf-mls-pq-ciphersuites.html
38. Amortized PQ MLS Combiner - https://messaginglayersecurity.rocks/mls-combiner/draft-ietf-mls-combiner.html - as of 2026-04-09
39. draft mahy mls x25519kyber768draft00 00 - https://www.ietf.org/archive/id/draft-mahy-mls-x25519kyber768draft00-00.txt - as of 2023-07-10

40. Direct Secure Messaging - DirectTrust - https://directtrust.org/what-we-do/direct-secure-messaging